

## THALA, Tunisia

WMO: 607380

Lat: 35.550N

Lon: 8.683E

Elev: 1092

StdP: 88.88

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.1       | 0.0 | -8.9                          | 2.0 | 11.9 | -6.8 | 2.4 | 10.9 | 18.9                  | 4.9  | 16.3 | 5.6  | 7.0                   | 320  | 0.588 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.5                   | 36.0            | 16.8 | 34.3 | 16.6 | 32.7 | 16.6 | 19.5                | 28.2 | 18.8 | 28.0 | 18.1 | 27.4 | 6.2                  | 320  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 17.3                            | 14.1 | 21.1 | 16.3 | 13.2 | 20.7 | 15.5 | 12.5 | 20.3 | 60.5          | 28.1 | 58.0 | 28.1 | 55.8 | 27.2 | 24.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 16.4          | 14.3 | 12.3 | -2.4                       | 38.8 | 1.4                | 1.1 | -3.4                                               | 39.6 | -4.2       | 40.2 | -5.0       | 40.8 | -6.0       | 41.6 |     |     |
| (5)               |      |      | -3.3                       | 21.1 | 1.5                | 1.3 | -4.4                                               | 22.0 | -5.3       | 22.8 | -6.2       | 23.5 | -7.3       | 24.5 | (4) | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 15.1  | 6.4   | 6.7   | 9.6   | 12.3  | 17.3  | 22.2  | 25.7  | 25.1  | 20.2  | 16.6  | 10.8  | 7.4  | (6)  |
| (7)  |                                                                                  | DBStd       | 7.78  | 3.25  | 3.56  | 4.21  | 3.91  | 4.35  | 4.34  | 3.48  | 3.33  | 3.60  | 3.75  | 3.41  | 3.26 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 427   | 121   | 104   | 58    | 19    | 1     | 0     | 0     | 0     | 1     | 29    | 93    |      | (8)  |
| (9)  |                                                                                  | HDD18.3     | 1905  | 371   | 326   | 271   | 185   | 73    | 13    | 1     | 1     | 20    | 81    | 226   | 339  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2283  | 8     | 11    | 48    | 88    | 228   | 367   | 488   | 467   | 307   | 204   | 54    | 12   | (10) |
| (11) |                                                                                  | CDD18.3     | 717   | 0     | 0     | 1     | 3     | 41    | 130   | 230   | 209   | 77    | 26    | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 7188  | 0     | 1     | 6     | 34    | 381   | 1354  | 2604  | 2138  | 531   | 139   | 1     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 3190  | 0     | 0     | 1     | 4     | 110   | 590   | 1308  | 992   | 158   | 27    | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 5.5   | 5.6   | 6.4   | 6.0   | 6.2   | 5.5   | 5.1   | 4.8   | 4.6   | 4.8   | 4.9   | 5.7   | 6.2  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 420   | 45    | 30    | 41    | 41    | 40    | 25    | 13    | 26    | 47    | 31    | 41    | 40   | (15) |
| (16) |                                                                                  | PrecMax     | 732   | 144   | 82    | 138   | 165   | 105   | 94    | 51    | 78    | 93    | 71    | 115   | 153  | (16) |
| (17) |                                                                                  | PrecMin     | 245   | 10    | 0     | 9     | 1     | 1     | 0     | 0     | 2     | 13    | 6     | 2     | 4    | (17) |
| (18) |                                                                                  | PrecStd     | 117   | 35    | 23    | 28    | 33    | 27    | 23    | 11    | 17    | 24    | 16    | 30    | 33   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 17.2  | 19.7  | 23.7  | 26.9  | 32.7  | 36.9  | 38.2  | 37.1  | 32.5  | 29.4  | 22.2  | 19.3 | (19) |
| (20) |                                                                                  |             | MCWB  | 8.4   | 9.4   | 11.3  | 13.0  | 15.3  | 16.2  | 17.0  | 17.3  | 16.7  | 14.7  | 10.3  | 9.7  | (20) |
| (21) |                                                                                  | 2%          | DB    | 14.8  | 16.4  | 20.8  | 23.8  | 29.7  | 34.3  | 36.5  | 35.3  | 30.4  | 26.7  | 19.6  | 16.2 | (21) |
| (22) |                                                                                  |             | MCWB  | 7.5   | 8.2   | 10.1  | 12.2  | 14.5  | 15.9  | 16.9  | 17.0  | 16.4  | 14.6  | 10.6  | 8.7  | (22) |
| (23) |                                                                                  | 5%          | DB    | 12.8  | 14.3  | 18.5  | 21.5  | 27.4  | 32.3  | 34.8  | 33.6  | 28.7  | 24.5  | 17.6  | 14.0 | (23) |
| (24) |                                                                                  |             | MCWB  | 6.6   | 7.6   | 9.5   | 11.6  | 14.4  | 15.9  | 16.8  | 17.0  | 16.4  | 13.8  | 10.1  | 8.0  | (24) |
| (25) |                                                                                  | 10%         | DB    | 11.1  | 12.2  | 16.5  | 19.4  | 25.3  | 30.2  | 33.1  | 31.9  | 26.7  | 22.4  | 15.8  | 12.2 | (25) |
| (26) |                                                                                  |             | MCWB  | 5.9   | 6.5   | 9.0   | 10.9  | 13.8  | 15.6  | 16.6  | 16.8  | 16.0  | 13.3  | 9.5   | 7.3  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 10.0  | 10.9  | 13.4  | 15.0  | 17.3  | 18.9  | 20.2  | 20.8  | 19.6  | 17.7  | 13.7  | 12.0 | (27) |
| (28) |                                                                                  |             | MCDWB | 13.8  | 16.6  | 18.3  | 20.8  | 25.9  | 29.2  | 29.4  | 28.2  | 25.3  | 21.7  | 17.7  | 16.3 | (28) |
| (29) |                                                                                  | 2%          | WB    | 8.7   | 9.3   | 11.7  | 13.7  | 16.3  | 17.8  | 19.2  | 19.7  | 18.5  | 16.3  | 12.5  | 10.1 | (29) |
| (30) |                                                                                  |             | MCDWB | 12.5  | 14.7  | 18.0  | 20.1  | 25.8  | 28.3  | 29.7  | 28.1  | 25.1  | 21.1  | 16.1  | 14.0 | (30) |
| (31) |                                                                                  | 5%          | WB    | 7.8   | 8.1   | 10.6  | 12.6  | 15.4  | 17.1  | 18.5  | 19.0  | 17.7  | 15.5  | 11.6  | 9.0  | (31) |
| (32) |                                                                                  |             | MCDWB | 11.3  | 13.0  | 16.7  | 19.2  | 24.6  | 27.5  | 29.3  | 28.0  | 24.2  | 20.7  | 14.8  | 12.1 | (32) |
| (33) |                                                                                  | 10%         | WB    | 6.8   | 7.1   | 9.6   | 11.7  | 14.5  | 16.4  | 17.8  | 18.3  | 17.0  | 14.8  | 10.7  | 8.2  | (33) |
| (34) |                                                                                  |             | MCDWB | 9.8   | 11.2  | 15.3  | 18.0  | 23.1  | 26.9  | 28.7  | 27.3  | 23.6  | 19.9  | 13.9  | 10.9 | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.1   | 7.0   | 8.6   | 9.5   | 10.9  | 11.8  | 12.5  | 11.6  | 9.7   | 8.1   | 6.7   | 5.8  | (35) |
| (36) |                                                                                  |             | MCDBR | 8.5   | 9.8   | 11.3  | 12.7  | 12.9  | 13.0  | 13.5  | 12.8  | 11.5  | 10.2  | 8.8   | 8.3  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.7   | 5.1   | 5.1   | 5.3   | 4.7   | 4.0   | 4.0   | 3.8   | 3.6   | 3.7   | 4.3   | 4.8  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 7.6   | 8.8   | 10.5  | 11.0  | 11.9  | 12.3  | 12.7  | 12.1  | 10.2  | 8.5   | 7.5   | 7.0  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.8   | 5.1   | 5.4   | 5.3   | 5.1   | 4.5   | 4.5   | 4.3   | 4.1   | 4.0   | 4.2   | 4.4  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.284 | 0.302 | 0.361 | 0.401 | 0.430 | 0.411 | 0.437 | 0.425 | 0.410 | 0.368 | 0.321 | 0.291 | (40) |      |
| (41) |                                                                                  | taud        | 2.488 | 2.435 | 2.257 | 2.154 | 2.093 | 2.181 | 2.106 | 2.138 | 2.192 | 2.288 | 2.417 | 2.485 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 926   | 943   | 905   | 883   | 858   | 872   | 845   | 849   | 843   | 850   | 867   | 893   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 82    | 98    | 128   | 149   | 161   | 148   | 158   | 150   | 135   | 112   | 88    | 78    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.70  | 3.49  | 4.68  | 5.77  | 6.64  | 7.50  | 7.52  | 6.72  | 5.18  | 4.01  | 2.95  | 2.41  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.22  | 0.24  | 0.31  | 0.38  | 0.35  | 0.33  | 0.22  | 0.31  | 0.23  | 0.19  | 0.18  | 0.20  | (45) |      |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (11 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page